

An Approach to a Case of Cough: Chronic Obstructive Pulmonary Disease (COPD)

Dr. Essam Kotb, Associate Professor, Internal Medicine | Source: COPD_ - .pdf (34 slides) · Refs: Step-Up 5th ed.; Harrison 20th ed.; AMBOSS; Cecil 10th ed.; First Aid IM Boards 4th ed.

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from the Pulmonary Vascular lecture (same lecturer):** That lecture listed **chronic lung disease (COPD)** as a cause of **Group 3 pulmonary hypertension** via **hypoxic vasoconstriction**, with **O₂ for COPD patients with PaO₂ < 60**. This lecture covers the disease itself, and you'll see the same chain here (hypoxia → pulmonary HTN → cor pulmonale).

COPD in one page: confirm → grade → group → treat

① CONFIRM (spirometry)

Post-bronchodilator
FEV1/FVC < 70% = obstruction
↑ TLC, ↑ FRC, ↑ RV (air trapping)
Emphysema: ↓ DLCO
Screen AAT (esp. young)

② GRADE severity (FEV1 %)

GOLD 1 mild ≥ 80%
GOLD 2 moderate 50–79%
GOLD 3 severe 30–49%
GOLD 4 very severe ... < 30%

③ GROUP (risk + symptoms)

**E: ≥ 2 exacerbations or
≥ 1 hospitalisation / yr**
Otherwise, by symptoms:
A: mMRC 0–1 and CAT < 10
B: mMRC ≥ 2 or CAT ≥ 10

④ TREAT by group

All: stop smoking · flu/pneumococcal/COVID vaccines · rehab · inhaler technique · SABA as needed

Group A

LAMA + SABA prn
(preferred)
or LABA / SAMA-SABA

Group B

LAMA + LABA
+ SABA prn

Group E

LAMA + LABA
+ SABA prn

Group E + eos ≥ 300 or hospitalisation

ICS + LAMA + LABA
+ SABA prn

AECOPD: O₂ to SpO₂ 88–92% · nebulised SABA ± SAMA · systemic steroids · antibiotics if purulent/ventilated

NIPPV if PaCO₂ ≥ 45 + pH ≤ 7.35, severe dyspnoea/fatigue, or persistent hypoxaemia

Intubate if NIPPV contraindicated or failing, confusion, impending arrest, haemodynamic instability

Added diagram: COPD from diagnosis to treatment

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Define COPD, chronic bronchitis, emphysema ; epidemiology	The "3 months × 2 years" definition
Knowledge	Exogenous and endogenous risk factors	Smoking ~90%; α1-antitrypsin
Knowledge	★ Mechanisms: inflammation, mucus, fibrosis, alveolar destruction, elastic recoil loss, air trapping, V/Q mismatch	Pathogenesis flowchart
Knowledge	★ Clinical features	Pink puffer vs blue bloater; no clubbing
Knowledge	★ GOLD 1–4 (FEV1%) and Groups A, B, E (mMRC/CAT + exacerbations)	Grading/grouping calculations

Domain	Objective	Likely tested as
Knowledge	★ Non-drug measures, stable maintenance (SABA, LABA, LAMA, ICS), AECOPD (O ₂ targets, steroids, bronchodilators, antibiotics, NIPPV)	Treatment by group; NIPPV indications
Skills	Differentiate phenotypes; interpret spirometry, ABG, imaging ; calculate mMRC ; plan stable and acute treatment	Case vignettes
Values	Smoking-cessation counselling, inhaler technique, compliance, empathy	

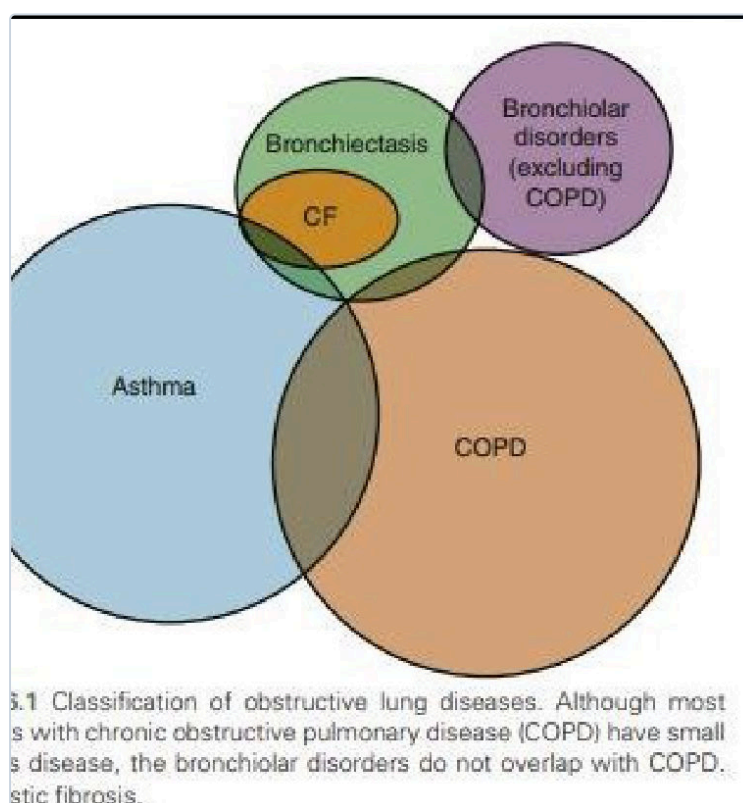
Flow: Definition → Aetiology → Pathophysiology → GOLD classification → Clinical features → Types (pink puffer / blue bloater) → Diagnostics → Management → Acute exacerbation → Summary → Cases.

2. Main Content (Lecture Order)

2.1 Introduction & Definitions (Slides 5-6)

★ **Obstructive lung disease / COPD:** persistent respiratory **symptoms (cough, dyspnoea)** + **airflow limitation**, caused by a combination of ★ **small-airway obstruction** and ★ **parenchymal destruction**.

Term	★ Definition
★ Chronic bronchitis	Productive cough ≥ 3 months per year for 2 consecutive years , not explained by another diagnosis (a clinical definition)
★ Emphysema	Permanent dilatation of air spaces distal to the terminal bronchioles from destruction of alveolar walls and pulmonary capillaries (a pathological definition)



Slide 5: Overlap of asthma, COPD, bronchiectasis, bronchiolar disorders, cystic fibrosis

Epidemiology (summary slide 29): ★ **M:F 3:2** · US prevalence ~6% · ★ **third most common cause of death worldwide** · **no curative therapy**.

🔄 **Recap: Definitions.** COPD = persistent symptoms + airflow limitation from small-airway disease + parenchymal destruction. Chronic bronchitis is clinical (cough 3 months × 2 years); emphysema is pathological (destroyed alveoli distal to the terminal bronchiole).

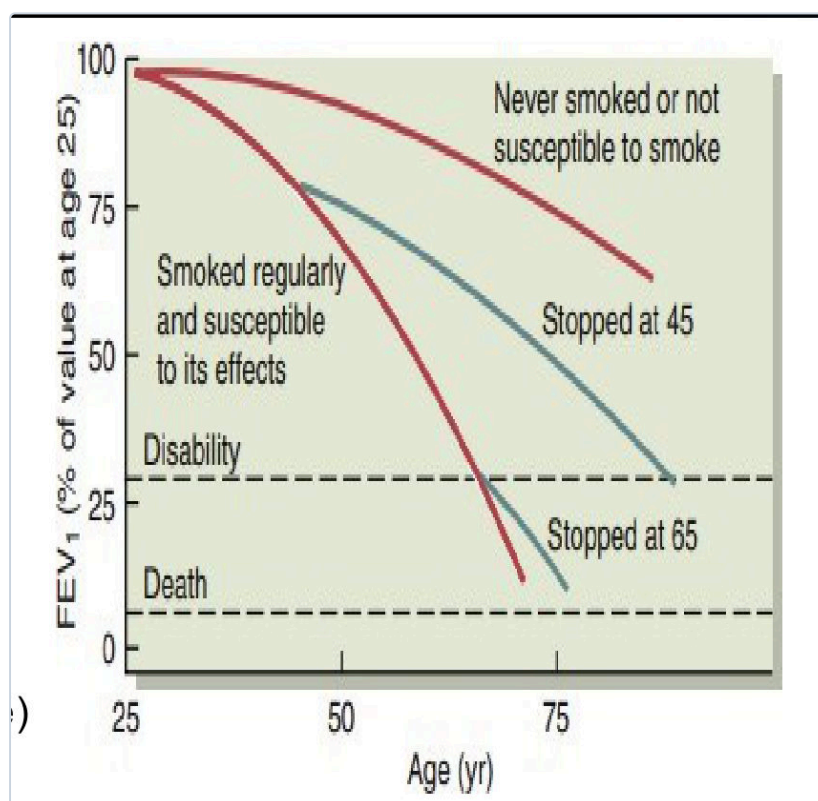
🔗 **Link:** Next, what causes it.

▶ **Up next: Aetiology.**

2.2 Aetiology (Slide 7)

★ Exogenous	Endogenous
★ Tobacco smoke (~90% of cases)	★ α1-antitrypsin deficiency
Air pollution / fine dusts (occupational)	Lung growth & development abnormalities: recurrent pulmonary infections and TB, premature birth
	Airway hyperresponsiveness
	Antibody deficiency (e.g., IgA deficiency)
	Primary ciliary dyskinesia (e.g., Kartagener syndrome)

★ **Fletcher-Peto curve (Slide 7 graph):** FEV₁ declines faster in **susceptible smokers; stopping smoking slows the decline** (stopping at 45 helps far more than stopping at 65) [curve name added].



Slide 7: FEV₁ decline with age: never-smoker vs susceptible smoker vs stopping at 45 or 65

🔄 **Recap: Aetiology.** Smoking ~90%; also pollution/dust. Endogenous: α 1-AT deficiency, abnormal lung development (infections, TB, prematurity), hyperresponsiveness, IgA deficiency, ciliary dyskinesia. Quitting earlier saves more FEV₁.

🔗 **Link:** These insults trigger two processes, inflammation and destruction.

▶ **Up next: Pathophysiology.**

2.3 Pathophysiology (Slides 8-9)

A. ★ Chronic inflammation (→ chronic bronchitis)

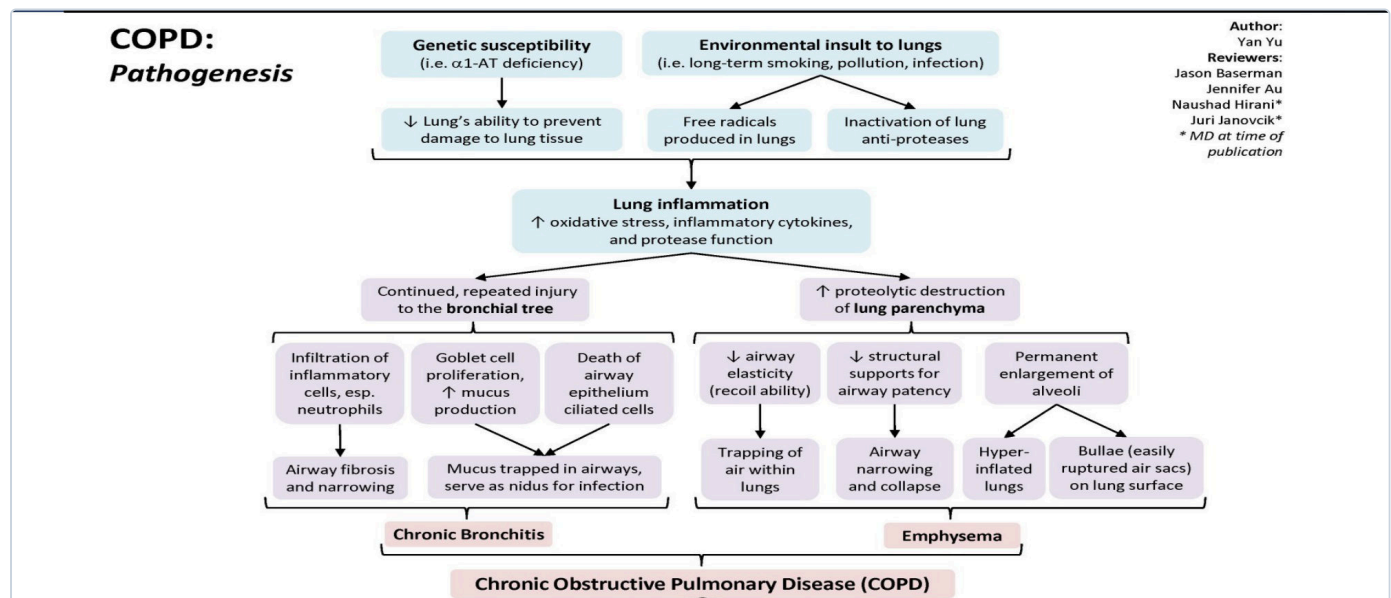
- ↑ **Neutrophils** → cytokine release
- ↑ **Growth factors** → **airway narrowing**
- ★ **Goblet-cell proliferation/hypertrophy** → **mucus hypersecretion** → **chronic productive cough**
- ★ **Hypoxic pulmonary vasoconstriction** → **pulmonary hypertension**

B. ★ Tissue destruction (→ emphysema)

- ★ **Nicotine inactivates protease inhibitors (especially α 1-antitrypsin)** → **protease-antiprotease imbalance** → **loss of elastic tissue**
- → **Enlarged airspaces** → ★ **↓ DLCO** → **hypoxaemia and hypercapnia**

★★ Pathogenesis flowchart (Slide 9):

1. **Genetic susceptibility** (α 1-AT deficiency) → ↓ lung's ability to prevent damage
2. **Environmental insult** (smoking, pollution, infection) → **free radicals** + **inactivated anti-proteases**
3. → **Lung inflammation**: ↑ oxidative stress, cytokines, protease activity
4. Two branches:
 - **Bronchial tree injury** → neutrophil infiltration → **fibrosis and narrowing** · goblet cells/↑ mucus + death of ciliated cells → **trapped mucus = nidus for infection** → ★ **CHRONIC BRONCHITIS**
 - **Proteolytic destruction of parenchyma** → ↓ elastic recoil → **air trapping** · ↓ structural support → **airway collapse** · permanent alveolar enlargement → **hyperinflation, bullae** → ★ **EMPHYSEMA**
5. Both → **COPD**



Slide 9: COPD pathogenesis: inflammation → chronic bronchitis; proteolysis → emphysema

🔗 **Recap: Mechanism.** Inflammation (neutrophils, goblet cells, mucus, narrowing) = chronic bronchitis. Protease excess (nicotine blocks α 1-AT) destroys elastin = emphysema (air trapping, $\downarrow$ DLCO). Hypoxic vasoconstriction $\rightarrow$ pulmonary HTN.

🔗 **Link:** Damage is measured by airflow (FEV1) and its impact on the patient (symptoms, exacerbations), which is GOLD.

👉 **Up next: GOLD classification.**

2.4 GOLD Classification (Slides 10-11)

☆☆☆ **Spirometric grade (post-bronchodilator FEV1 % predicted):**

Grade	FEV1
GOLD 1: mild	$\geq 80\%$
GOLD 2: moderate	50-79%
GOLD 3: severe	30-49%
GOLD 4: very severe	$< 30\%$
<i>Mnemonic [added]: "80-50-30".</i>	

☆☆☆ **GOLD groups (exacerbation risk + symptoms):**

Group	Exacerbations in past year	mMRC	CAT
A	0, or 1 not leading to admission	0-1	< 10 (0-9)
B	0, or 1 not leading to admission	≥ 2	≥ 10
★ E	★ ≥ 2, or ≥ 1 leading to hospital admission	Any	Any

★ **mMRC dyspnoea scale (Slide 24 table):**

Grade	Breathless...
0	Only with strenuous exercise
1	Hurrying on level ground or walking up a slight hill
2	Walks slower than peers on level ground, or stops for breath at own pace
3	Stops after ~ 100 yards or a few minutes on level ground
4	Too breathless to leave the house / breathless when dressing

🔗 **Recap: GOLD.** Grades 1-4 by FEV1 (80/50/30). Groups: E = ≥ 2 exacerbations or ≥ 1 admission; otherwise A (mMRC 0-1, CAT < 10) or B (mMRC ≥ 2 or CAT ≥ 10).

🔗 **Link:** Now what the patient looks like at the bedside.

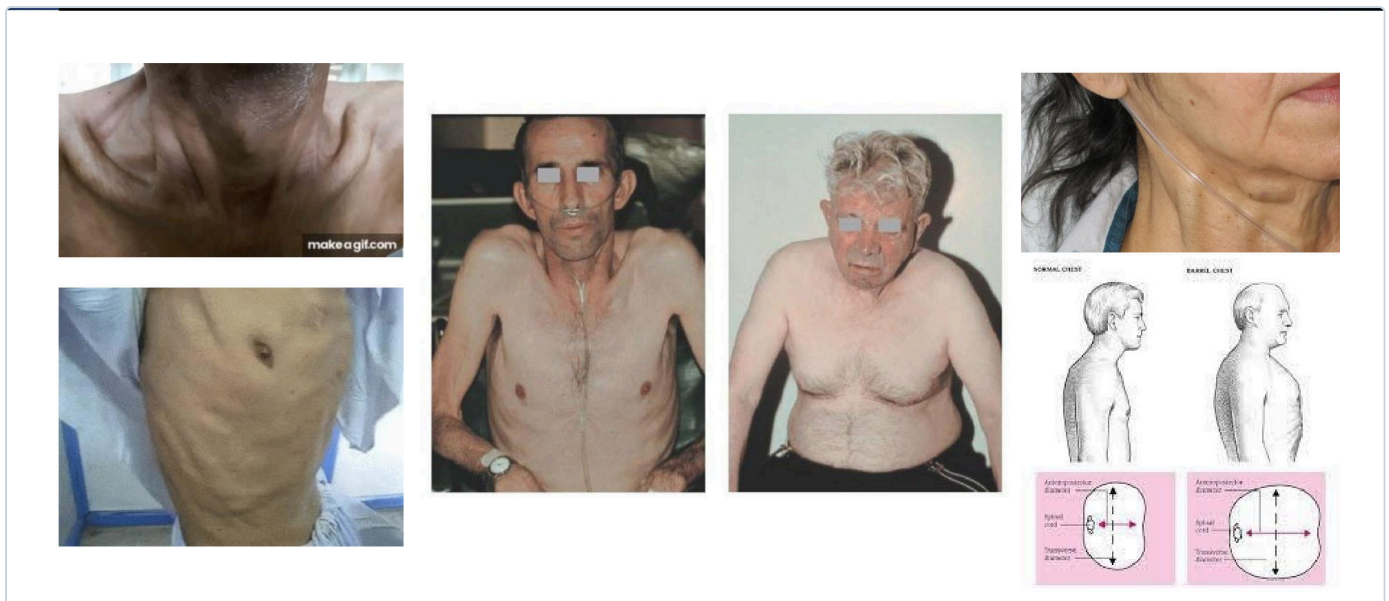
👉 **Up next: Clinical features.**

2.5 Clinical Features (Slides 12-14)

★ **Symptoms are minimal/non-specific until advanced.**

- **Chronic cough, prolonged expiration, dyspnoea, wheezing**
- **Chest hyperinflation; distant breath and heart sounds**
- ★ **Clubbing is NOT seen in COPD** [added: if present, look for lung cancer, bronchiectasis or ILD]
- **Accessory-muscle use**

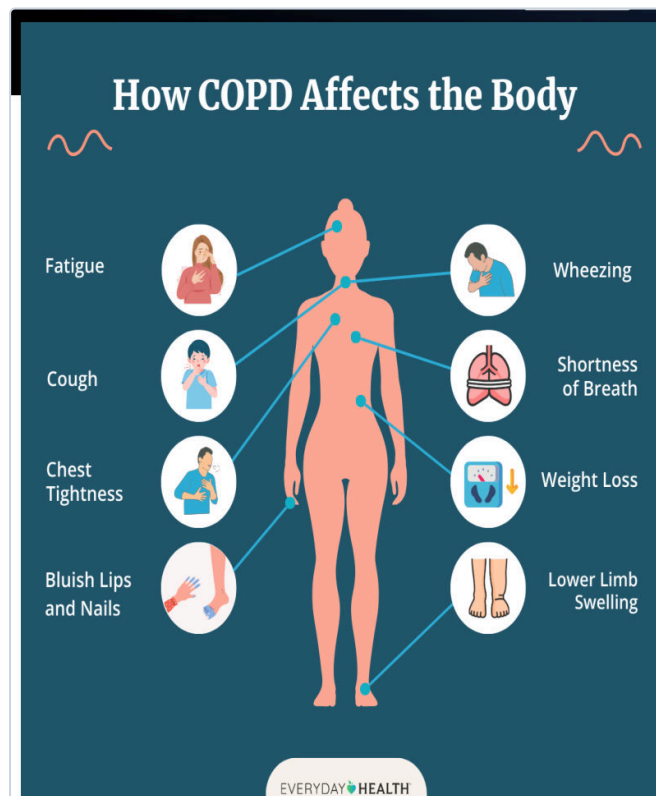
- **Cyanosis** ("blue bloater"); **pursed-lip breathing** ("pink puffer")



Slide 13: Accessory muscles, cachexia (pink puffer), plethora/oedema (blue bloater), barrel chest with ↑ AP diameter

★★ Advanced COPD (Slide 14):

- **Congested neck veins**
- ★ **Barrel chest**: most common in **emphysema**
- **Asynchronous chest/abdomen movement**; accessory muscles (diaphragm dysfunction)
- ★ **Hyperresonance**, ↓ **diaphragmatic excursion**, ↓ **cardiac dullness**
- ★ ↓ **breath sounds** ("silent lung")
- **Peripheral oedema**, ★ **RV hypertrophy & cor pulmonale**, **hepatomegaly**
- **Weight loss, cachexia**
- ★ **Secondary polycythaemia**
- **Confusion** (hypoxaemia, hypercapnia)

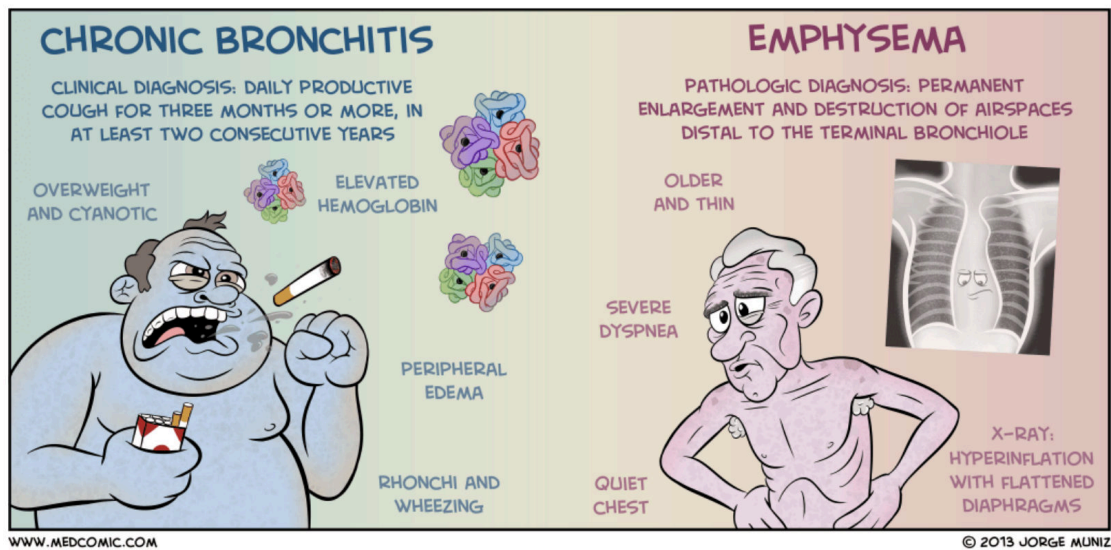


Slide 14: How COPD affects the body

2.6 Types: Pink Puffer vs Blue Bloater (Slides 15-16)

★ Usually **hard to separate**; a **combination usually exists**.

	★ Pink puffer	★ Blue bloater
Main mechanism	Emphysema	Chronic bronchitis
Look	Non-cyanotic , ★ cachectic , ★ pursed-lip breathing	★ Cyanosed , overweight , ★ peripheral oedema
Cough	Mild	★ Productive
PaO ₂	Slightly ↓	★ Markedly ↓
PaCO ₂	Normal (hypercapnia only late)	★ ↑ (early hypercapnia)



Slide 16: Chronic bronchitis vs emphysema: "COPD comes in two flavours"

🔄 **Recap: Clinical.** Cough, dyspnoea, wheeze, hyperinflation, no clubbing. Advanced: barrel chest, hyperresonance, silent chest, cor pulmonale, polycythaemia, cachexia, confusion. Pink puffer (emphysema: thin, pursed lips, near-normal gases) vs blue bloater (bronchitis: cyanosed, oedematous, ↑ CO₂).

🔗 **Link:** Suspicion is confirmed with spirometry; imaging and ABG support it.

📌 **Up next: Diagnostics.**

2.7 Diagnostics (Slides 17-21)

History (smoking / noxious gases) + **examination** + **investigations:**

Test	★ Findings
★ CXR	↑ AP diameter, flattened diaphragm, horizontal ("transverse") ribs, long narrow heart, widened intercostal spaces, hyperlucent lungs
Chest CT	Similar to CXR (emphysematous spaces, bullae)
★ Serum α1-antitrypsin	Screen all patients, especially the young
ABG	★ Chronic CO ₂ retention, ↓ PO ₂

Test	★ Findings
★★ Spirometry (PFT)	★ FEV1/FVC < 70%, ↓ FEV1, ↑ TLC, ↑ FRC, ↑ RV ; "scooped-out" expiratory flow-volume curve



Anterior (left) and lateral (right) chest radiographs show changes caused by severe emphysema, including hyperinflation, increased anterior-posterior diameter, flattened pulmonary arteries.

Slide 18: CXR: hyperinflation, flat diaphragm, ↑ AP diameter

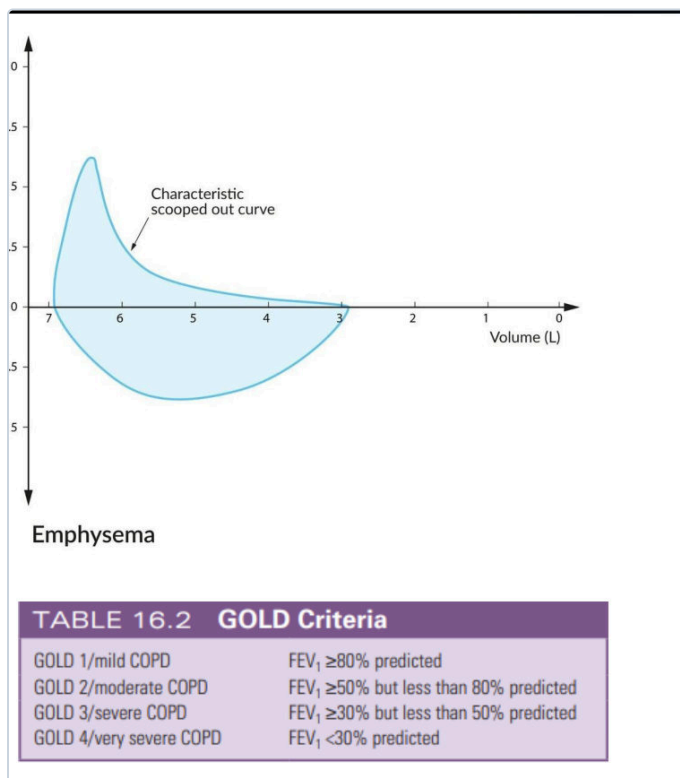


E-Fig. 16.7 Computed tomographic scan shows centrilobular/paraseptal emphysema.

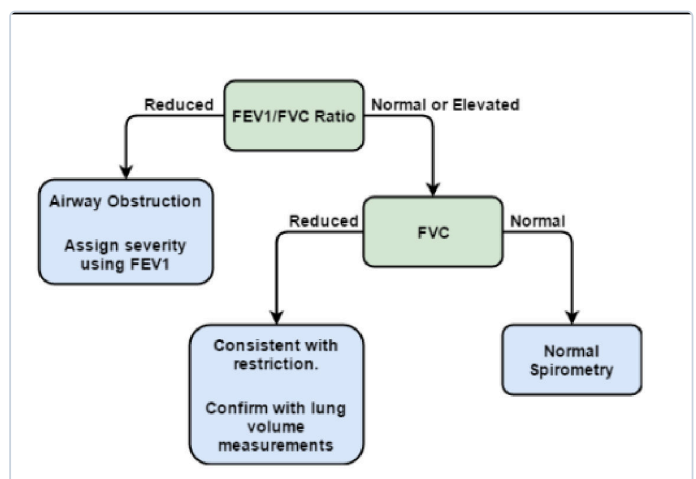
Slide 19: CT: centrilobular/paraseptal emphysema

★★ Spirometry interpretation (Slide 21):

1. **FEV1/FVC reduced** → ★ **obstruction** → **grade by FEV1**
2. **FEV1/FVC normal/high** → look at **FVC**:
 - **FVC reduced** → **restriction**, confirm with lung volumes
 - **FVC normal** → **normal spirometry**



Slide 20: Scooped-out flow-volume loop in emphysema; GOLD criteria



Slide 21: Spirometry interpretation algorithm

🔗 **Recap: Diagnosis.** FEV1/FVC < 70% confirms obstruction; FEV1 grades it. Volumes ↑ (TLC, FRC, RV). CXR hyperinflation. ABG ↑ CO₂ ↓ O₂. Check α1-AT, especially in the young.

🔗 **Link:** With the grade and group known, treatment follows the GOLD group.

🔗 **Up next: Management of stable COPD.**

2.8 Management of Stable COPD (Slides 22-24)

Approach (Slide 22):

1. Assess **symptoms + exacerbation history** → **GOLD group**; and **comorbidities** (e.g., asthma)
2. ★ **Non-pharmacological care for everyone**
3. ★ **Drugs based on GOLD group**
4. Follow-up: ★ **review inhaler technique**, adjust to symptoms, ★ **annual spirometry** (FEV1 decline)
5. Advanced disease: **long-term oxygen, interventional treatment**

★ **Non-pharmacological:** ★ **smoking cessation** · ★ **vaccines: influenza, pneumococcal (+ COVID-19 per algorithm)** · **pulmonary rehab** · regular physical activity

★ Pharmacological (Slide 23):

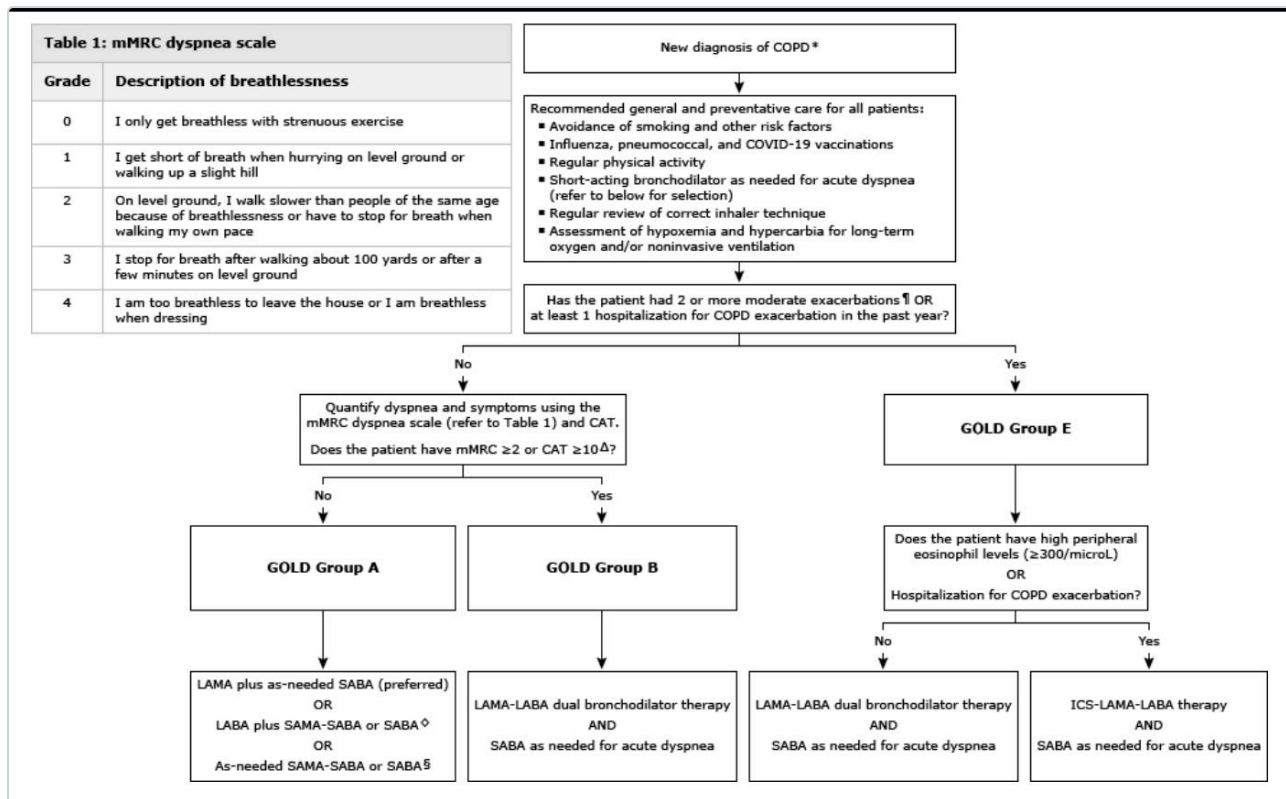
- ★ **Oxygen if SpO₂ ≤ 88% at rest**
- ★ **Inhaled muscarinic antagonists = main line**
- Inhaled corticosteroids · inhaled β₂-agonists · pulmonary rehab
- **Mucolytics** (N-acetylcysteine, erdosteine): break **disulfide bonds** of mucoproteins; may ↓ exacerbations in some patients
- **Additional:** surgery, antibiotics, **PDE-4 inhibitors**, opiates, theophylline

★ Inhaler table (Slide 23):

Duration	LABA	LAMA	Inhaled steroids
12 h	Salmeterol, formoterol	Acclidinium, darotroprum	Beclomethasone, budesonide, fluticasone propionate
24 h	Indacaterol, vilanterol, olodaterol, abediterol, carmoterol	★ Tiotropium , glycopyrronium, umeclidinium	Ciclesonide, mometasone, fluticasone furoate
<i>Mnemonic [added]: *"-terol" = β-agonist; "-ium" = antimuscarinic; "-sone/-nide" = steroid.</i>			

★★★★ Treatment by GOLD group (Slide 24 algorithm):

Group	Initial therapy (+ SABA as needed for acute dyspnoea)
A	★ LAMA + as-needed SABA (preferred) ; or LABA + SAMA-SABA/SABA; or as-needed SAMA-SABA/SABA
B	★ LAMA-LABA dual bronchodilator
E (eos < 300, no admission)	LAMA-LABA
E with ★ eosinophils ≥ 300/μL or hospitalisation for exacerbation	★ ICS-LAMA-LABA (triple)



Slide 24: mMRC scale and GOLD A/B/E treatment algorithm

Recap: Stable. Everyone: quit smoking, vaccines, rehab, inhaler check, SABA prn. A → LAMA; B and E → LAMA + LABA; E with eosinophils ≥ 300 or admission → add ICS (triple). O₂ if SpO₂ $\leq 88\%$ at rest. Repeat spirometry yearly.

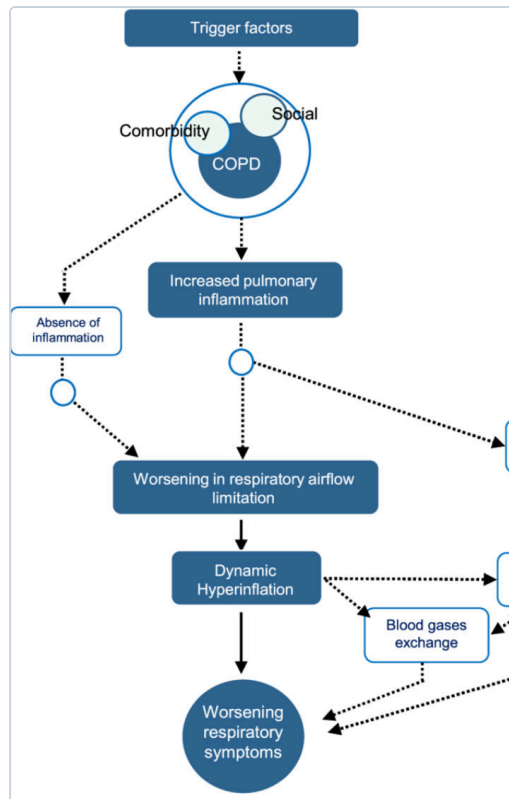
Link: Even well-managed patients flare. Exacerbations drive decline and admissions.

Up next: Acute exacerbation.

2.9 Acute Exacerbation of COPD, AECOPD (Slides 25-28)

★ **Definition:** worsening dyspnoea, ↑ cough, and a change in sputum volume or purulence. **Cause:** ★ viral or bacterial infection (pneumonia). **Investigations:** CXR, ABG, flu swab, sputum culture.

Mechanism (Slide 25): trigger → ↑ pulmonary (± systemic) inflammation → **worse airflow limitation** → ★ **dynamic hyperinflation** → worse gas exchange and cardiovascular strain → **worsening symptoms**.



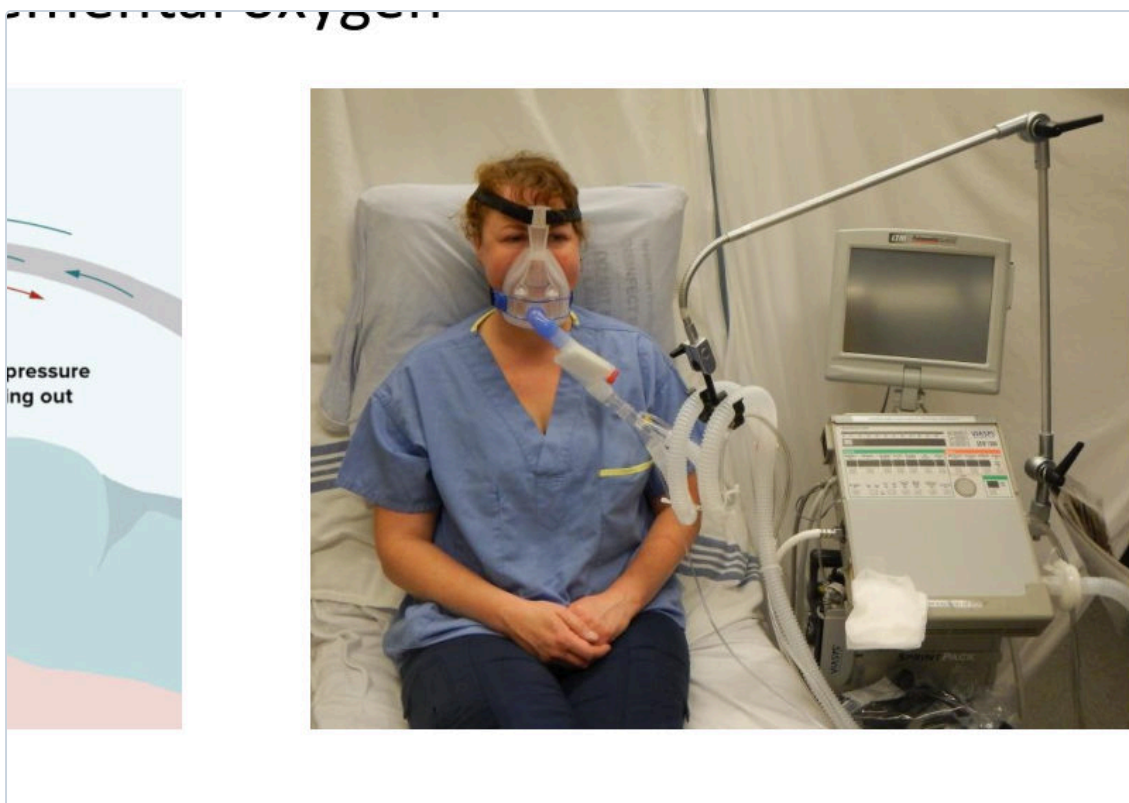
Slide 25: Exacerbation cascade: inflammation → airflow limitation → dynamic hyperinflation → symptoms

★★★★ Hospital management (Slide 26):

1. ★ **O₂ to SpO₂ 88-92%** [added: higher targets risk worsening hypercapnia]
2. ★ **Bronchodilators:** nebulised **SABA ± SAMA** (± ICS per slide)
3. ★ **Systemic glucocorticoids**
4. ★ **Antibiotics if indicated: profuse purulent sputum or needing ventilation**
5. Adjuncts if indicated (e.g., **diuretics** for pulmonary hypertension)
6. Watch for **impending respiratory failure** (cyanosis, tachypnoea) → **ICU**
7. ★ **Serial ABGs**; if hypercapnia or persistent hypoxaemia, **titrate O₂ to target**

★★★★ NIPPV indications (Slide 27):

- ★ **Respiratory acidosis: PaCO₂ ≥ 45 mm Hg and pH ≤ 7.35**
- **Severe dyspnoea** with ↑ work of breathing / **respiratory-muscle fatigue**
- **Persistent hypoxaemia despite O₂**



Slide 27: Non-invasive positive pressure ventilation

★★ Intubation indications (Slide 28):

- **NIPPV contraindicated: facial trauma, repeated vomiting, uncooperative patient**
- **Confusion**
- ★ **Persistent respiratory acidosis despite NIPPV**
- **Impending respiratory arrest**
- **Haemodynamic instability**

🔄 **Recap: AECOPD.** Worse dyspnoea + cough + sputum change, usually infection. O_2 88–92%, nebulised SABA/SAMA, systemic steroids, antibiotics if purulent or ventilated, serial ABGs. NIPPV if $pH \leq 7.35$ with $PaCO_2 \geq 45$, fatigue or refractory hypoxaemia. Intubate if NIPPV fails or can't be used, or with confusion, arrest risk or shock.

2.10 Summary (Slide 29)

COPD
Chronic obstructive pulmonary disease

Epidemiology
Sex: 3:2 male/female ratio
Prevalence (US): ~6%
Third most common cause of death worldwide

Etiology/risk factors
Exogenous: tobacco smoking, air pollution, occupational exposures
Endogenous: α 1-Antitrypsin deficiency, antibody deficiency syndromes, premature birth

Diagnosis
Pulmonary function tests:
Blood gas analysis
Imaging: chest x-ray, chest CT
 α 1-Antitrypsin if suspecting AATD

Complications
Acute exacerbations (AECOPD)
Chronic respiratory failure

Note
No curative therapy

Slide 29: COPD summary card and clinical picture

- **Epidemiology:** M:F **3:2**, prevalence ~6%, **3rd cause of death worldwide**
- **Risk:** smoking, pollution, occupation; α 1-AT deficiency, antibody deficiency, prematurity
- **Diagnosis:** PFT, ABG, CXR/CT, **α 1-AT if AATD suspected**
- **Complications:** **AECOPD, chronic respiratory failure** (and right heart failure on the figure)
- **No curative therapy**

3. High-Yield Comparison Tables

Table A: Chronic bronchitis vs emphysema

	Chronic bronchitis	Emphysema
Definition type	Clinical (cough 3 mo $\times$ 2 y)	Pathological (destruction distal to terminal bronchiole)
Mechanism	Inflammation, goblet cells, mucus	Protease > antiprotease, elastin loss
Phenotype	Blue bloater	Pink puffer
DLCO	Normal-ish [added]	★ ↓
CO ₂	↑ early	Normal until late
Cor pulmonale	★ Early/common	Late

Table B: GOLD grades and groups (see §2.4)

Table C: Stable vs acute treatment

	Stable	Exacerbation
O ₂	If SpO ₂ $\leq$ 88% at rest	Target 88-92%

	Stable	Exacerbation
Bronchodilator	Long-acting (LAMA ± LABA) + SABA prn	Nebulised short-acting (SABA ± SAMA)
Steroid	Inhaled (only if eos ≥ 300 / admitted)	Systemic
Antibiotic	Selected cases	If purulent sputum / ventilated
Ventilation	Long-term NIV in selected patients	NIPPV → intubation

Table D: NIPPV vs intubation

NIPPV	Intubation
pH ≤ 7.35 + PaCO ₂ ≥ 45 · severe dyspnoea/fatigue · persistent hypoxaemia	NIPPV contraindicated (facial trauma, vomiting, uncooperative) · confusion · acidosis despite NIPPV · impending arrest · haemodynamic instability

Table E: Obstructive vs restrictive spirometry *[restrictive column partly added]*

	Obstructive (COPD)	Restrictive
FEV1/FVC	↓ (< 70%)	Normal or ↑
FVC	Normal or ↓	↓
TLC, RV	↑	↓

4. Rules, Exceptions, Patterns

- **Spirometry confirms COPD: post-bronchodilator FEV1/FVC < 70%.**
- **FEV1 grades severity (80/50/30); exacerbations + symptoms decide the group.**
- **One hospital admission alone puts the patient in Group E.**
- **LAMA is the main line;** add LABA for B/E; add ICS only for eosinophils ≥ 300 or hospitalisation.
- **No clubbing in COPD.**
- **Emphysema: ↓ DLCO.** Asthma/pure bronchitis: DLCO normal or ↑ *[added]*.
- **O₂ stable: give if SpO₂ ≤ 88%. O₂ in AECOPD: target 88-92%** (avoid over-oxygenation).
- **Systemic steroids in exacerbations, inhaled steroids in selected stable patients.**
- **Screen α1-AT in all, especially the young.**
- **Smoking cessation is the only measure that clearly slows FEV1 decline** (Fletcher-Peto) *[added emphasis]*.

5. Clinical Correlations

- **Young patient (< 45) with basal emphysema, little smoking → α1-antitrypsin deficiency** *[basal and liver-disease link added]*.
- **Blue bloater with ankle oedema, raised JVP → cor pulmonale** (Group 3 PH, see the Pulmonary Vascular lecture).
- **Secondary polycythaemia** from chronic hypoxaemia.
- **Confusion and drowsiness in AECOPD** → CO₂ narcosis → check ABG; NIPPV/intubation.
- **Infertility + situs inversus + bronchiectasis** → Kartagener syndrome (ciliary dyskinesia).
- **Clubbing in a COPD patient** → look for lung cancer *[added]*.
- **Shipyard work** → asbestos and dust exposure (Case 1).

6. Rapid Revision (Last-Day)

- COPD = persistent symptoms + airflow limitation from small-airway disease + parenchymal destruction.
- **Chronic bronchitis: productive cough ≥ 3 mo/yr $\times$ 2 consecutive yrs. Emphysema: destroyed alveoli distal to terminal bronchiole.**
- M:F 3:2, ~6%, **3rd cause of death**, no cure.
- **Smoking ~90%**; pollution/dust. Endogenous: **$\alpha 1$ -AT**, prematurity, TB/infections, hyperresponsiveness, **IgA deficiency, Kartagener.**
- Inflammation: neutrophils, goblet cells $\rightarrow$ mucus $\rightarrow$ cough; hypoxic vasoconstriction $\rightarrow$ pulmonary HTN.
- Destruction: **nicotine inactivates $\alpha 1$ -AT** $\rightarrow$ elastin loss $\rightarrow$ **$\downarrow$ DLCO**, hypoxaemia, hypercapnia.
- **GOLD 1 $\geq 80 \cdot 2$ 50-79 $\cdot$ 3 30-49 $\cdot$ 4 < 30 .**
- **A: ≤ 1 non-admitted exacerbation, mMRC 0-1, CAT < 10 . B: mMRC ≥ 2 / CAT ≥ 10 . E: ≥ 2 exacerbations or ≥ 1 admission.**
- Signs: cough, prolonged expiration, wheeze, hyperinflation, **no clubbing**, accessory muscles, pursed lips, cyanosis.
- Advanced: barrel chest, hyperresonance, silent lung, cor pulmonale, hepatomegaly, polycythaemia, cachexia, confusion.
- **Pink puffer = emphysema (thin, pursed lips, near-normal gases). Blue bloater = bronchitis (cyanosed, oedema, early $\uparrow$ CO₂).**
- CXR: $\uparrow$ AP diameter, flat diaphragm, horizontal ribs, narrow heart, hyperlucency.
- **PFT: FEV1/FVC $< 70\%$, $\downarrow$ FEV1, $\uparrow$ TLC/FRC/RV.** $\alpha 1$ -AT screen. ABG $\uparrow$ CO₂ $\downarrow$ O₂.
- Stable: quit smoking, flu + pneumococcal vaccines, rehab, inhaler technique, annual spirometry; **O₂ if SpO₂ $\leq 88\%$; LAMA = main line.**
- **A: LAMA + SABA. B: LAMA + LABA. E: LAMA + LABA $\rightarrow$ + ICS if eos ≥ 300 or admitted.**
- AECOPD: dyspnoea + cough + sputum change; infection. **O₂ 88-92%**, nebulised SABA $\pm$ SAMA, **systemic steroids**, antibiotics if purulent/ventilated, serial ABG.
- **NIPPV: pH ≤ 7.35 + PaCO₂ ≥ 45 , fatigue, refractory hypoxaemia. Intubate:** NIPPV contraindicated or failing, confusion, arrest, shock.

7. Exam-Style Questions

Q1 (MCQ, lecture Case 1, Slide 30). 68 ♂, 1 year of increasing dyspnoea and productive cough; ex-shipyard worker; **30 pack-years**, quit 16 years ago; childhood asthma. SpO₂ 93%, scattered wheeze and rhonchi. **FEV1/FVC 63%, FEV1 65%, ★ DLCO 40%**. Most likely cause? A) Asthma B) Chronic bronchitis C) CHF D) Pulmonary HTN E) Bronchiectasis **F) Emphysema Answer: F.** [reasoning added; the slide gives no key] Obstruction (ratio $< 70\%$, GOLD 2 by FEV1 65%) with a **markedly reduced DLCO** = alveolar-capillary destruction = **emphysema**. Asthma and chronic bronchitis typically have **normal DLCO**. Pulmonary HTN doesn't cause obstruction. ⚠ He has a productive cough (bronchitis feature), but DLCO is the discriminator the question is built on.

Q2 (MCQ, lecture Case 2, Slide 31). 65 ♀ with COPD on salmeterol: 2 days of worse dyspnoea, cough, **yellow sputum**; HR 104, RR 28, **SpO₂ 85%**, wheeze, CXR hyperinflation without consolidation; on nasal O₂. Most appropriate initial drug therapy? A) Prednisone + roflumilast B) Albuterol + montelukast C) Prednisone + salmeterol D) Albuterol + theophylline **E) Prednisone + albuterol F) Prednisone + tiotropium Answer: E.** [reasoning added] AECOPD $\rightarrow$ **short-acting bronchodilator (SABA) + systemic corticosteroid**. Long-acting agents (salmeterol, tiotropium) and roflumilast are maintenance drugs. Montelukast and theophylline aren't first-line. Antibiotics would also be reasonable given the sputum colour change, but weren't offered as an option.

Q3 (MCQ). Spirometry FEV1/FVC 0.62, FEV1 42% predicted. GOLD grade? a) 1 b) 2 **c) 3** d) 4 **Answer: c** (30–49%).

Q4 (MCQ). A COPD patient had **one hospital admission** for an exacerbation last year; mMRC 1, CAT 8. GOLD group? a) A b) B **c) E** d) Can't classify **Answer: c.** ≥ 1 admission = E regardless of symptoms.

Q5 (Short answer). List the NIPPV indications in AECOPD and three reasons to intubate instead.
Answer: NIPPV: **respiratory acidosis ($\text{PaCO}_2 \geq 45$ mm Hg and $\text{pH} \leq 7.35$), severe dyspnoea with fatigue / $\uparrow$ work of breathing, persistent hypoxaemia despite O_2 .** Intubate: **NIPPV contraindicated** (facial trauma, repeated vomiting, uncooperative), **confusion, persistent acidosis despite NIPPV, impending arrest, haemodynamic instability.**

8. Exam Pitfalls & Traps

- Chronic bronchitis is defined by spirometry.** No, it's **clinical** (cough 3 mo $\times$ 2 y); emphysema is **pathological**.
- Clubbing in COPD.** Not a feature; look for another cause.
- Group by FEV1.** Groups use **exacerbations + mMRC/CAT**; FEV1 gives the **grade** only.
- Group E needs 2 exacerbations. Or one hospitalisation.**
- ICS for everyone.** Only Group E with **eosinophils ≥ 300** or admission (triple therapy).
- High-flow O_2 to 98% in AECOPD.** Target **88–92%**.
- Long-acting bronchodilators for an acute attack.** Use **short-acting** (nebulised SABA $\pm$ SAMA) + **systemic steroids**.
- Low DLCO = bronchitis. Low DLCO = emphysema.**
- Pink puffer is cyanosed.** No, the **blue bloater** is.
- ⚠ Slide inconsistency:** the stable-management bullet says "**inhaled muscarinic antagonists (main line)**" and lists ICS before β_2 -agonists, but the GOLD algorithm adds ICS only for eosinophils ≥ 300 / admission. Follow the algorithm.
- ⚠ AECOPD slide** lists "nebulised SBA, SAB/SAMA & ICS"; read as **SABA $\pm$ SAMA**; systemic steroids are the key steroid in exacerbations.
- ⚠ Slide typos:** "thiopheline" = **theophylline**; "fascial trauma" = **facial trauma**; "darotropium" appears in the LAMA list.